

HSC Practice Questions

Budgeting

- 1 Mary is 18 years old and has just purchased comprehensive motor vehicle insurance. The following excesses apply to claims for at-fault motor vehicle accidents.

- Basic excess of \$850 for each claim
- An additional age excess of \$1600 for drivers under 25 years of age
- An additional age excess of \$400 for drivers 25 years of age or over with no more than 2 years driving experience

How much would Mary be required to pay as excess if she made a claim as the driver at fault in a car accident?

- A \$1600 B \$850 + \$400
C \$850 + \$1600 D \$850 + \$1600 + \$400

(1 mark)

(06, 2019 HSC) Easy

- 2 The driving distance from Alex's home to his work is 20 km. He drives to and from work five times each week. His car uses fuel at the rate of 8 L /100 km.

How much fuel does he use driving to and from work each week?

- A 16 L B 20 L
C 25 L D 40 L

(1 mark)

(05, 2018 HSC) Easy

- 3 Sam is the driver at fault in a car accident.

Which of the following is covered by Sam's compulsory third-party (CTP) insurance?

- A Repairs to Sam's car
B Injury to the other driver
C Damage to the other driver's car
D Cost of repairing a building damaged in the accident

(1 mark)

(Q15, 2018 HSC) Medium

- 4 David earns a gross income of \$890 per week. Each week, 25% of this income is deducted in taxation. David budgets to save 20% of his net income.

How much does he budget to save each week?

- A \$44.50 B \$133.50
C \$489.50 D \$534.00

(1 mark)

(Q21, 2018 HSC) Easy

- 5 Kate is comparing two different models of car. Car A uses fuel at the rate of 9 L /100 km. Car B uses 3.5 L /100 km.

Suppose Kate plans on driving 8000 km in the next year.

How much less fuel will she use driving car B instead of car A?

- A 280 L B 440 L
C 720 L D 1000 L

(1 mark)

(Q14, 2017 HSC) Easy

- 6 An old washing machine uses 130 L of water per load. A new washing machine uses 50 L per load.

How much water is saved each year if two loads of washing are done each week using the new machine?

- A 2600 L B 4160 L
C 5200 L D 8320 L

(1 mark)

(Q9, 2016 HSC) Easy

- 7 Alice intends to buy a car and insure it.

Briefly describe what each of these types of insurance covers:

- Compulsory third-party insurance (CTP)
- Non-compulsory third-party property insurance.

(2 marks)

(Q27a, 2016 HSC) Medium

- 8 An insurance company offers customers the following discounts on the basic annual premium for car insurance.

Type of discount	Discount
Multi-policy discount (Owner has more than 2 insurance policies with the company)	15%
No-claim bonus (Owner has had at least 5 years without an insurance claim)	20%
Combined CTP and comprehensive insurance (Owner has both insurances with the company)	\$50

If a customer is eligible for more than one discount, subsequent discounts are applied to the already discounted premium. The combined compulsory third party (CTP) and comprehensive insurance discount is always applied last.

Jamie has three insurance policies, including combined CTP and comprehensive insurance, with this company. He has used this company for 8 years and he has never made a claim.

The basic annual premium for his car insurance is \$870.

How much will Jamie need to pay after the discounts are applied?

- A \$482.44 B \$515.50
C \$541.60 D \$557.60 (1 mark)

(Q25, 2015 HSC) Medium

- 9 Alex is buying a used car which has a sale price of \$13 380. In addition to the sale price there are the following costs:

Transfer of registration \$30
Stamp Duty

- i Stamp Duty for this car is calculated at \$3 for every \$100, or part thereof, of the sale price.
Calculate the Stamp Duty payable. (1 mark) Medium
- ii Alex borrows the total amount to be paid for the car including Stamp Duty and transfer of registration. Interest on the loan is charged at a flat rate of 7.5% per annum. The loan is to be repaid in equal monthly instalments over 3 years.
Calculate Alex's monthly repayments. (4 marks) Medium
- iii Alex wishes to take out comprehensive insurance for the car for 12 months. The cost of comprehensive insurance is calculated using the following:

Base rate \$845
Fire Service
Levy (FSL) 1% of base rate
Stamp Duty 5.5% of the total of base rate and FSL
GST 10% of the total of base rate and FSL.

Find the total amount that Alex will need to pay for comprehensive insurance. (3 marks) Easy

- iv Alex has decided he will take out the comprehensive car insurance rather than the less expensive non-compulsory third-party car insurance. What extra cover is provided by the comprehensive car insurance? (1 mark) Medium

(Q27a, 2014 HSC)

- 10 Xuso is comparing the costs of two different ways of travelling to university.

Xuso's motorcycle uses one litre of fuel for every 17 km travelled. The cost of fuel is \$1.67/L and the distance from her home to the university car park is 34 km. The cost of travelling by bus is \$36.40 for 10 single trips.

Which way of travelling is cheaper and by how much? Support your answer with calculations. (2 marks)

(Q27b, 2014 HSC) Easy

- 11 A section of Jim's electricity bill is shown.

Energy Used and Costs					
METER ID	THIS READING	LAST READING	ENERGY USED	RATE (per kWh)	COST
Peak Energy Charge					
TMV04221/01	531.2	274.8	256.4 kWh	47.7700c	\$122.48
Shoulder Energy Charge					
TMV04221/02	A	560.9	523.5 kWh	19.4000c	\$101.56
Off-peak (Night) Energy Charge					
TMV04221/03	242.5	0.0	242.5 kWh	9.6000c	\$23.28

- i What is the value of A? (1 mark) Medium
- ii How much will Jim save if he uses 154 kWh of energy at the Off-peak rate rather than at the Peak rate? (2 marks) Medium

(Q26d, 2013 HSC)

- 12 Warrick has a net income of \$590 per week. He has created a budget to help manage his money.

	A	B	C
1	Warrick's Weekly Budget		
2			
3	Rent	\$175	
4	Petrol	\$45	
5	Car registration	\$10	
6	Car maintenance	\$15	
7	Food	\$90	
8	Electricity	X	
9	Telephone and internet	\$40	
10	Insurances	\$30	
11	Entertainment	\$70	
12	Clothes and gifts	\$50	
13	Savings	\$40	
14	Total	\$590	

- i Find the value of X, the amount that Warrick allocates towards electricity each week. (1 mark) Easy
- ii Warrick has an unexpectedly high telephone and internet bill. For the last three weeks, he has put aside his savings as well as his telephone and internet money to pay the bill.
How much money has he put aside altogether to pay the bill? (1 mark) Easy
- iii The bill for the telephone and internet is \$620. It is due in two weeks time. Warrick realises he has not put aside enough money to pay the bill.

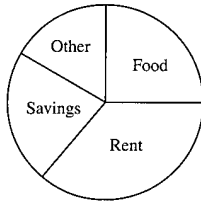
Budgeting

How could Warrick reallocate non-essential funds in his budget so he has enough money to pay the bill? Justify your answer with suitable reasons and calculations.

(3 marks) **Medium**

(Q23d, 2010 HSC)

- 13 Margaret has a weekly income of \$900 and allocates her money according to the budget shown in the sector graph.



How long will it take Margaret to save \$3600?

- A 4 weeks B 5 weeks
C 16 weeks D 18 weeks

(1 mark)

(Q7, 2007 HSC) **Medium**

- 14 Reece is preparing his annual budget for 2006. His expected income is:

- \$90 every week as a swimming coach
- Interest earned from an investment of \$5000 at a rate of 4% per annum.

His planned expenses are:

- \$30 every week on transport
- \$12 every week on lunches
- \$48 every month on entertainment.

Reece will save his remaining income. He uses the spreadsheet below for his budget.

	A	B	C	D	E	F	G	H
1	REECE'S ANNUAL BUDGET FOR 2006							
2								
3	INCOME				EXPENSES			
4								
5	Wages			\$4,680		Transport	\$(Y)	
6	Interest on investment			\$(X)		Lunches	\$624	
7						Entertainment	\$(Z)	
8								
9								

- i Determine the values of (X), (Y) and (Z). (Assume there are exactly 52 weeks in a year.) (3 marks) **Medium**
- ii At the beginning of 2006, Reece starts saving. Will Reece have saved enough money during 2006 for a deposit of \$2100 on a car if he keeps to his budget? Justify your answer with suitable calculations. (2 marks) **Medium**

(Q25a, 2005 HSC)

Budgeting

- 1 Mary must pay the basic excess, plus the age excess because she is under 25 years.
She must pay $\$850 + \1600 .
Answer C

- 2 Total distance = $5 \times 2 \times 20 \text{ km}$
= 200 km
At 8 L/100 km 16 L of fuel will be used.
Answer A

- 3 Injury to the other driver is covered by CTP insurance.
Answer B

- 4 Net income = 75% of $\$890$
= $\$667.50$
Budgeted savings
= 20% of $\$667.50$
= $\$133.50$
Answer B

- 5 $8000 \text{ km} = 80 \times 100 \text{ km}$
Car A uses 9 L/100 km.
Fuel used by car A = $80 \times 9 \text{ L}$
= 720 L
Car B uses 3.5 L/100 km.
Fuel used by car B = $80 \times 3.5 \text{ L}$
= 280 L
Difference = $(720 - 280) \text{ L}$
= 440 L
Answer B

- 6 Saving per load = $(130 - 50) \text{ L}$
= 80 L
Yearly savings = $2 \times 52 \times 80 \text{ L}$
= 8320 L
Answer D

- 7 CTP insurance covers injuries to passengers and other people in an accident. ✓
Third-party property insurance covers damage to other people's cars and property. ✓
(2 marks)

- 8 Multi-policy discount
= 15% of $\$870$
= $\$130.50$
Price after discount
= $\$870 - \130.50
= $\$739.50$
No-claim bonus discount
= 20% of $\$739.50$
= $\$147.90$
Price after this discount
= $\$739.50 - \147.90
= $\$591.60$

Combined insurance discount = $\$50$
Final price
= $\$591.60 - \50
= $\$541.60$
Answer C

- 9 i $\$13\,380 = 133 \times \$100 + \$80$
So there will be 134 lots of \$3 charged.
Stamp duty = $134 \times \$3$
= $\$402$
(1 mark)
ii Total borrowed
= $\$13\,380 + \$402 + \$30$
= $\$13\,812$ ✓
 $I = Prn$
= $\$13\,812 \times 0.075 \times 3$
= $\$3107.70$ ✓
Total to repay
= $\$13\,812 + \3107.70
= $\$16\,919.70$ ✓
Now 3 years = $3 \times 12 \text{ months}$
= 36 months

Each repayment
= $\$16\,919.70 \div 36$
= $\$469.991\,666\dots$
= $\$469.99$ (nearest cent) ✓
(4 marks)

- iii Base rate = $\$845$
FSL = 1% of $\$845$
= $\$8.45$ ✓
Base rate + FSL
= $\$845 + \8.45
= $\$853.45$

Stamp duty = 5.5% of $\$853.45$
= $0.055 \times \$853.45$
= $\$46.939\,75$
= $\$46.94$ ✓
(nearest cent)

GST = 10% of $\$853.45$
= $\$85.345$
= $\$85.35$ (nearest cent)
Total = $\$853.45 + \46.94
+ $\$85.35$
= $\$985.74$ ✓ ✓
(3 marks)

- iv Comprehensive car insurance covers damage or theft of the car of the person taking out the insurance whereas (noncompulsory) third party car insurance

only covers damage done by that vehicle to other people's cars and property.
(1 mark)

- 10 Fuel used in single trip
= $(34 \div 17) \text{ L}$
= 2 L

Cost of fuel = $2 \times \$1.67$
= $\$3.34$ ✓

Cost of single bus trip
= $\$36.40 \div 10$
= $\$3.64$

Difference = $\$3.64 + \3.34
= $\$0.30$

So, based on fuel costs alone, it is 30 cents cheaper for each single trip to travel by motorcycle. ✓
[But this ignores all the other costs associated with owning and operating the motorcycle, so is not a true picture of the difference in costs.]
(2 marks)

- 11 i $A = 560.9 + 523.5$
= 1084.4
(1 mark)

- ii Difference in rates
= $(47.77 - 9.6) \text{ c/kWh}$
= 38.17 c/kWh ✓
For 154 kW:
Savings = $154 \times 38.17\text{c}$
= 5878.18c
= $\$58.78$ ✓
[nearest cent]
(2 marks)

- 12 i Total shown in spreadsheet
= $\$(175 + 45 + 10 + 15 + 90 + 40 + 30 + 70 + 50 + 40)$
= $\$565$
 $X = \$590 - \565
= $\$25$
(1 mark)

- ii Amount put aside each week
= $\$40 + \40
= $\$80$
Total for 3 weeks = $3 \times \$80$
= $\$240$
(1 mark)

Budgeting

iii Amount still required

$$= \$620 - \$240$$

$$= \$380$$

Amount from 2 more weeks of savings and phone and internet funds

$$= 2 \times \$80$$

$$= \$160$$

Extra amount required

$$= \$380 - \$160$$

$$= \$220$$

Extra amount required per week

$$= \$220 \div 2$$

$$= \$110$$

Non-essential items are entertainment and clothing and gifts.

$$\text{Total of these} = \$70 + \$50$$

$$= \$120$$

Possible answers could show how Warrick could find the extra amount needed from entertainment and clothing and gifts funds.

(3 marks)

13 Angle size of savings = 80°

(by measurement)

$\therefore$ Amount saved per week

$$= \frac{80}{360} \times \$900$$

$$= \$200$$

Time taken to save \$3600

$$= \$3600 \div \$200$$

$$= 18 \text{ weeks}$$

Answer D

14 i X = interest on investment

$$= \frac{4}{100} \times \$5000$$

$$= \$200$$

Y = annual transport expenses

$$= 52 \times \$30$$

$$= \$1560$$

Z = annual entertainment

expenses

$$= 12 \times \$48$$

$$= \$576$$

(3 marks)

ii Total income

$$= \$4680 + \$200$$

$$= \$4880$$

Total expenses

$$= \$1560 + \$624 + \$576$$

$$= \$2760$$

Savings

$$= \$4880 - \$2760$$

$$= \$2120$$

Yes. Reece will be able to save the \$2100 deposit for his car.

(2 marks)